

```

n = [4 3 9];
d = [4 3 -4];
% a) Pole and Zero plot
zplane(n, d);
title('Pole-Zero Plot');
grid on;
% b) Partial
[r, p, k] = residuez(n, d);
disp('Residues:'), disp(r);
disp('Poles:'), disp(p);
disp('Direct Term:'), disp(k);

```

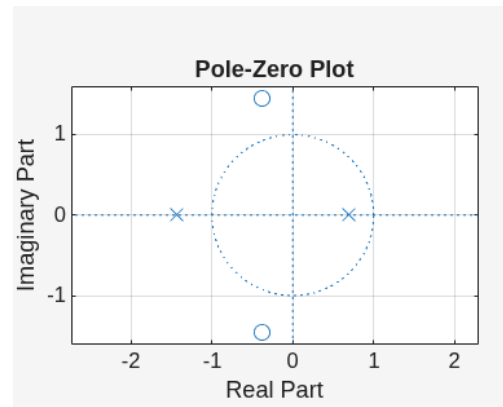
```

Residues:
    1.0544
    2.1956

Poles:
   -1.4430
    0.6930

Direct Term:
   -2.2500

```



```

z = [0.21; 3.14; -0.3 + 0.5j; -
0.3 - 0.5j];
p = [-0.45; -0.67; 0.81 +
0.72j; 0.81 - 0.72j];
k = 2.2;
[b, a] = zp2tf(z, p, k)
H = tf(b, a, 1) % 1 for z-t

```

$$H = \frac{2.2 z^4 - 6.05 z^3 - 2.223 z^2 - 1.635 z + 0.4932}{z^4 - 0.5 z^3 - 0.3384 z^2 + 0.827 z + 0.3541}$$

```

syms z
num = 18*z^3;
den = 18*z^3 + 3*z^2 - 4*z - 1;
G = num / den;
[G_partial] = pretty (partfrac(G, z))

```

$$G_{\text{partial}} = \frac{9}{25(2z - 1)} - \frac{26}{25(3z + 1)} + \frac{2}{5(3z + 1)^2} + 1$$

```

n = [1 -1 0]; % [z^2 z^1 z^0] for
z(z-1)
d = [1 4 3]; % [z^2 z^1 z^0] for
(z+1)(z+3)
% Partial fraction expansion
[R, P, K] = residuez(n, d);
% Display results
disp('Residues (R):'), disp(R);
disp('Poles (P):'), disp(P);
disp('Direct Term (K):'), disp(K);

```

```

Residues (R):
    2.0000
   -1.0000

Poles (P):
   -3
   -1

Direct Term (K):
    0

```